100

Name Brandon Buchanan Quiz 4 1/25/2007

Let f be the mod-7 function.

1) What is f(35)? 0

2) What is the GCD of 100 and 75? 25

3) What is the LCM of 100 and 75? 300

4) What is 211 in base 5? 1321

5) What is 211 in base 2? 11010011

6) What is 0110_2 in base 10? 6

7) 27 mod 5 = 2, so 27 = 5 * 5 + 2

8) If n mod 4 = 2, then n = 0 + 2 for some n?

9) If n | m, then m (mod n) = 0 and m = k n for some k?

10) Prove If a (mod 3) = 2 and b (mod 3) = 1, then a + b (mod 3) = 0. $xk + n = y$

assume a (mod 3) = 2 and b (mod 3) = 1,

so a + b (mod 3) = 3

so 3 (mod 3) = 0

thus a + b (mod 3) = 0